Aurora Forecasting – Background Information and Additional Resources

Authors: Kira Lee, University of Tasmania

Background information:

- The aurora is a spectacular naturally occurring light display within Earth's upper atmosphere (SWS BoM).
  - ***Aurora borealis*** refers to the aurora in the northern hemisphere (the northern lights) (NASA, 2025).
  - ***Aurora australis*** refers to the aurora in the southern hemisphere (the southern lights) (NASA, 2025).

- The aurora has fascinated humanity for millennia and features in the historical records of many cultures:
  - In 10th century Chinese literature, it is described as a **"five-coloured light"** (van der Sluijs, 2023).
  - In Aboriginal Australian oral tradition, it features strongly as an omen of bloodshed, death and disease and has been described as **"bushfires into the spirit world"** (Hamacher, 2013).

- Auroras occur when the sun releases high energy charged particles in an event called a **Coronal Mass Ejection (CME)** (SWS BoM). CMEs create **geomagnetic storms** which allow these solar particles to enter Earth's magnetosphere. The solar particles are funnelled towards the poles, where they interact with gases in Earth's atmosphere. These interactions release light, which is seen as the aurora.

- Auroras are concentrated in ring-shaped regions called **auroral ovals** at the poles. The auroral oval extends towards the equator during larger storm events (SWPC NOAA).

- The **colour** of the aurora depends on which gases in the atmosphere are hit by incoming solar particles, and the altitude that this occurs (NASA, 2025).
  - **Oxygen** produces green between 100-200 km altitude, and red above 200 km.
  - **Nitrogen** produces pink below 100 km altitude, and blue between 100-200 km.
  - Colours can mix to make others including purples, pinks and white.

- The aurora can take many different forms including arcs, bands, coronas and diffuse glows. Auroral form can evolve over time (NASA, 2008).

- The visibility of the aurora depends on three factors – the solar wind speed, Kp index and Bz index (69Nord, 2025).
  - **Solar wind speed (km/s):** the speed of the stream of charged particles emitted from the sun. The faster the solar wind speed, the more likely it is to see the aurora.
  - **Kp index:** a measure of Earth's geomagnetic activity on a scale from 0-9. The higher the value, the greater the chance of seeing the aurora.
  - **Bz (nT):** describes the orientation of the solar wind's own magnetic field. A negative (southward) value means the aurora is more likely as the orientation of the solar wind's own magnetic field is opposite to Earth's magnetic field. This allows for more interaction between them (SWPC NOAA).

- Space weather parameters like solar wind speed are measured by space-based instruments like NASA's **DSCOVR** (Deep Space Climate Observatory). This satellite is located between the Earth and the Sun at a point called the L1 (Lagrange) point around 1.5 million kilometers from Earth (NOAA NESDIS). Real-time data from DSCOVR can be found here.

- To see an aurora, you need dark, clear skies away from sources of light pollution. Peak viewing hours are generally between 10pm – 2am. To see a bright aurora, the moon should not be full. You should find a location that faces north in the northern hemisphere or south in the southern hemisphere (SWPC NOAA).

References:

69Nord (2025). *Understanding the indicators of Northern Lights viewing apps.* Available at: https://69nord.com/en/announcement/understanding-the-indicators-of-northern-lights-viewing-apps/

Bureau of Meteorology Australian Space Weather Forecasting Centre (SWS BoM). *Aurora Photos and the Aurora Explained.* Available at: https://www.sws.bom.gov.au/Educational/1/1/4

Hamacher, D. (2013). Aurorae in Australian Aboriginal Traditions, *Journal of Astronomical History & Heritage*, 16(2), pp. 207-219.

NASA (2008). *Magnetic Mysteries of the Aurora.* Available at: https://cse.ssl.berkeley.edu/SegwayEd/lessons/exploring_magnetism/space_weather/index.html

NASA (2025). *Auroras*. Available at: https://science.nasa.gov/sun/auroras/

National Environmental Satellite, Data, and Information Service (NOAA NESDIS). *DSCOVR: Deep Space Climate Observatory.* Available at: https://www.nesdis.noaa.gov/our-satellites/currently-flying/dscovr-deep-space-climate-observatory

Space Weather Prediction Center National Oceanic and Atmospheric Administration (SWPC NOAA). *Space Weather Glossary*. Available at: https://www.swpc.noaa.gov/content/space-weather-glossary

Space Weather Prediction Center National Oceanic and Atmospheric Administration (SWPC NOAA). *Tips on Viewing the Aurora*. Available at: https://www.swpc.noaa.gov/content/tips-viewing-aurora

van der Sluijs, M. and Hayakawa, H. (2023). A candidate auroral report in the Bamboo Annals, indicating a possible extreme space weather event in the early 10th century BCE. *Advances in Space Research*. 72. 5767-5776. https://doi.org/10.1016/j.asr.2022.01.010

Additional resources:

Youtube videos:

ESA video of the aurora from space: European Space Agency (ESA) (2017). *Stunning aurora as seen from the Space Station*. Available at: https://youtu.be/XzBp6Bt503o?si=jdwKvAwmOFQPDV0l

Photographs:

A collection of aurora photographs sent to the Australian Space Weather Forecasting Centre: Bureau of Meteorology Australian Space Weather Forecasting Centre (SWS BoM). *Aurora Photos Sent to ASWFC*. Available at: https://www.sws.bom.gov.au/Educational/1/1/5

Webpages:

NASA Auroras Webpage: NASA (2025). *Auroras*. Available at: https://science.nasa.gov/sun/auroras/

NOAA Aurora Tutorial: Space Weather Prediction Center National Oceanic and Atmospheric Administration (SWPC NOAA). *Aurora Tutorial*. Available at: https://www.swpc.noaa.gov/content/aurora-tutorial

NOAA Space Weather Prediction Center tips on viewing the aurora: Space Weather Prediction Center National Oceanic and Atmospheric Administration (SWPC NOAA). *Tips on Viewing the Aurora*. Available at: https://www.swpc.noaa.gov/content/tips-viewing-aurora

Real Time Solar Wind Data: Space Weather Prediction Center National Oceanic and Atmospheric Administration (SWPC NOAA). *Real Time Solar Wind*. Available at: https://www.swpc.noaa.gov/products/real-time-solar-wind

Additional classroom activities:

Additional aurora-based classroom activities for high-school students, including a photo-based activity on different aurora forms (Activity 11): NASA (2008). *Magnetic Mysteries of the Aurora*. Available at: https://cse.ssl.berkeley.edu/SegwayEd/lessons/exploring_magnetism/space_weather/index.html

For tips on catching the aurora australis around Hobart, including a map of viewing locations: Strikis, A. (2026). *Complete Guide to the Southern Lights in Tasmania | Aurora Australis*. Available at: https://lapoftasmania.com.au/southern-lights-in-tasmania/